

Volatility forecasting with ECB news sentiment — tables

Auto-generated

Table 1: Descriptive statistics of returns and stance series.

Sample	n	Return mean (%)	Return SD (%)	Return skew	Return kurt	Realized vol (annu.)	P_t mean	N_t mean	S_t mean
In-sample (2008–2018)	2754	-0.0141	1.4894	-0.0627	5.6993	23.6442	0.3059	-0.2776	0.0283
OOS (2019–2023)	1260	0.0329	1.3270	-0.9683	13.0791	21.0658	0.3180	-0.3449	-0.0269
Full	4014	0.0007	1.4404	-0.2895	7.4236	22.8662	0.3097	-0.2987	0.0109

Table 2: In-sample MLE estimates, main suite. Lower AIC is better.

model	n_obs	loglik		aic	n_params
B1.1	2754	8193.0270	-16378.0550		4
B1.2	2754	8253.4470	-16496.8940		5
B1.3	2754	8270.7140	-16531.4290		5
NA1.1	2754	8193.1620	-16372.3250		7
NA1.2	2754	8194.9810	-16373.9630		8

Table 3: In-sample parameter estimates with std. errors in parentheses (main suite). ω, α, β are GARCH-family variance parameters; γ_{lev} is the GJR/EGARCH leverage term; a, b, κ, γ are the NA-GARCH news-scaling parameters; ν is the Student- t degrees of freedom. Std. errors for NA-GARCH variants are computed from the numerical Hessian with delta-method transforms back to natural parameter space. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ from the studentised estimate–standard-error ratio.

param	B1.1	B1.2	B1.3	NA1.1	NA1.2
omega	0.0000** (0.0000)	0.0000*** (0.0000)	-0.1752*** (0.0033)	0.0000*** (0.0000)	0.0000** (0.0000)
alpha	0.0833*** (0.0141)	0.0000 (0.0124)	0.1177*** (0.0172)	0.0835	0.0781
beta	0.9119*** (0.0141)	0.8965*** (0.0197)	0.9801*** (0.0054)	0.9113	0.9177
gamma_lev		0.1905*** (0.0337)	-0.1626*** (0.0205)		
a				1.0005*** (0.0061)	0.9829*** (0.0061)
b				0.0067	0.0923
kappa				0.0067	0.0860 (0.2469)
gamma					3.0120
nu	6.7690*** (0.8773)	7.5123*** (1.0806)	7.8666*** (1.1907)	6.7466*** (0.8534)	6.7116*** (0.8442)

Table 4: In-sample MLE estimates, robustness suite.

model	dist	n_obs	loglik	aic
R1.2	studentst	2749	8179.5970	-16343.1940
R1.3	studentst	2752	8188.9690	-16361.9380
R1.4	studentst	2754	8193.1650	-16370.3300
R1.5	studentst	2754	8195.2140	-16374.4280
R1.6	studentst	2754	8195.2860	-16374.5730
R1.7 (NA1.2 Gaussian)	normal	2754	8140.2400	-16266.4790

Table 5: Ljung-Box test on standardized residuals and squared standardized residuals (main suite). H_0 : no autocorrelation.

model	lag	Q_resid	p_resid	Q_resid_sq	p_resid_sq
B1.1	5	8.5410	0.1290	3.7410	0.5870
B1.1	10	11.8130	0.2980	6.4830	0.7730
B1.1	20	18.0530	0.5840	17.1260	0.6450
B1.2	5	8.2320	0.1440	6.8910	0.2290
B1.2	10	12.2680	0.2680	9.0050	0.5320
B1.2	20	18.5100	0.5540	18.4430	0.5580
B1.3	5	8.4810	0.1320	2.9920	0.7010
B1.3	10	12.7410	0.2390	4.0830	0.9440
B1.3	20	19.2890	0.5030	13.9870	0.8310
NA1.1	5	8.3170	0.1400	4.1300	0.5310
NA1.1	10	11.5920	0.3130	6.8220	0.7420
NA1.1	20	17.6550	0.6100	17.3890	0.6280
NA1.2	5	8.8420	0.1160	4.1480	0.5280
NA1.2	10	12.0760	0.2800	7.1850	0.7080

Table 6: Out-of-sample evaluation: QLIKE and RMSE by model, scheme (rolling window / increasing window) and horizon. Lower QLIKE is better.

model	scheme	horizon	n_obs	QLIKE	RMSE
B1.3	IW	1	252	-7.7734	0.0006
B1.2	IW	1	252	-7.7581	0.0006
NA1.1	IW	1	252	-7.6612	0.0006
B1.1	IW	1	252	-7.6609	0.0006
NA1.2	IW	1	252	-7.6384	0.0006
B1.3	IW	5	252	-6.3754	0.0016
B1.2	IW	5	252	-6.3653	0.0020
B1.1	IW	5	252	-6.3072	0.0018
NA1.1	IW	5	252	-6.3072	0.0018
NA1.2	IW	5	252	-6.2922	0.0018
B1.3	RW	1 ₆	252	-7.7672	0.0006
B1.2	RW	1	252	-7.7576	0.0006

Table 7: Diebold-Mariano tests of equal predictive accuracy. $\Delta_{QL} = \text{QLIKE}(\text{model}) - \text{QLIKE}(\text{baseline})$; negative values favour the candidate model. Newey-West HAC standard errors with bandwidth $h - 1$ for $h > 1$.

model	vs_baseline	scheme	horizon	mean_diff_QL	DM_stat	p_value	interp
B1.2	B1.1	RW	1	-0.1022	-2.1310	0.0331	model better
B1.2	B1.1	RW	5	-0.0571	-2.0294	0.0424	model better
B1.2	B1.1	IW	1	-0.0972	-2.0953	0.0361	model better
B1.2	B1.1	IW	5	-0.0581	-2.2349	0.0254	model better
B1.3	B1.1	RW	1	-0.1118	-1.7216	0.0851	no diff.
B1.3	B1.1	RW	5	-0.0654	-1.9530	0.0508	no diff.
B1.3	B1.1	IW	1	-0.1125	-1.8142	0.0696	no diff.
B1.3	B1.1	IW	5	-0.0682	-2.2037	0.0275	model better
NA1.1	B1.1	RW	1	0.0000	0.0583	0.9535	no diff.
NA1.1	B1.1	RW	5	-0.0004	-1.6114	0.1071	no diff.
NA1.1	B1.1	IW	1	-0.0004	-1.3717	0.1702	no diff.
NA1.1	B1.1	IW	5	0.0000	0.0987	0.9214	no diff.
NA1.2	B1.1	RW	1	0.0146	1.2333	0.2175	no diff.
NA1.2	B1.1	RW	5	0.0117	1.5242	0.1275	no diff.
NA1.2	B1.1	IW	1	0.0225	1.6233	0.1045	no diff.
NA1.2	B1.1	IW	5	0.0150	1.3987	0.1619	no diff.
R1.2	B1.1	RW	1	0.0012	0.8885	0.3743	no diff.
R1.2	B1.1	RW	5	0.0013	1.9164	0.0553	no diff.
R1.3	B1.1	RW	1	0.0168	1.2309	0.2184	no diff.
R1.3	B1.1	RW	5	0.0045	0.6177	0.5367	no diff.
R1.4	B1.1	RW	1	-0.0001	-0.2153	0.8295	no diff.
R1.4	B1.1	RW	5	-0.0002	-1.0001	0.3173	no diff.
R1.4	B1.2	RW	1	-0.1022	-2.1310	0.0331	model better
R1.4	B1.2	RW	5	-0.0571	-2.0294	0.0424	model better
R1.4	B1.2	IW	1	-0.0972	-2.0953	0.0361	model better
R1.4	B1.2	IW	5	-0.0581	-2.2349	0.0254	model better
R1.4	B1.3	RW	1	-0.1118	-1.7216	0.0851	no diff.
R1.4	B1.3	RW	5	-0.0654	-1.9530	0.0508	no diff.
R1.4	B1.3	IW	1	-0.1125	-1.8142	0.0696	no diff.
R1.4	B1.3	IW	5	-0.0682	-2.2037	0.0275	model better
R1.4	NA1.1	RW	1	0.0000	0.0583	0.9535	no diff.
R1.4	NA1.1	RW	5	-0.0004	-1.6114	0.1071	no diff.
R1.4	NA1.1	IW	1	-0.0004	-1.3717	0.1702	no diff.
R1.4	NA1.1	IW	5	0.0000	0.0987	0.9214	no diff.
R1.4	NA1.2	RW	1	0.0146	1.2333	0.2175	no diff.
R1.4	NA1.2	RW	5	0.0117	1.5242	0.1275	no diff.
R1.4	NA1.2	IW	1	0.0225	1.6233	0.1045	no diff.
R1.4	NA1.2	IW	5	0.0150	1.3987	0.1619	no diff.
R1.4	R1.2	RW	1	0.0012	0.8885	0.3743	no diff.
R1.4	R1.2	RW	5	0.0013	1.9164	0.0553	no diff.
R1.4	R1.3	RW	1	0.0168	1.2309	0.2184	no diff.
R1.4	R1.3	RW	5	0.0045	0.6177	0.5367	no diff.
R1.4	R1.4	RW	1	-0.0001	-0.2153	0.8295	no diff.
R1.4	R1.4	RW	5	-0.0002	-1.0001	0.3173	no diff.

Table 8: Hansen-Lunde-Nason (2011) Model Confidence Set at 90% confidence. Models in the MCS cannot be statistically eliminated as inferior.

scheme	horizon	n_obs	in_MCS	excl
RW	1	252	B1.2, B1.3	B1.1, NA1.1, NA1.2, R1.2, R1.3, R1.4
RW	5	252	B1.2, B1.3	B1.1, NA1.1, NA1.2, R1.2, R1.3, R1.4
IW	1	252	B1.2, B1.3	B1.1, NA1.1, NA1.2
IW	5	252	B1.2, B1.3	B1.1, NA1.1, NA1.2

Table 9: Sub-sample OOS evaluation across regimes: 2019 (normal), 2020–21 (COVID), 2022–23 (inflation shock).

subsample	model	scheme	horizon	n_obs	QLIKE	RMSE
2019__normal	B1.2	IW	1	50	-8.6880	0.0001
2019__normal	B1.3	IW	1	50	-8.6670	0.0001
2019__normal	NA1.2	IW	1	50	-8.5838	0.0001
2019__normal	NA1.1	IW	1	50	-8.5673	0.0001
2019__normal	B1.1	IW	1	50	-8.5673	0.0001
2019__normal	B1.3	IW	5	50	-7.0319	0.0004
2019__normal	B1.2	IW	5	50	-7.0274	0.0004
2019__normal	NA1.2	IW	5	50	-6.9681	0.0004
2019__normal	B1.1	IW	5	50	-6.9615	0.0004
2019__normal	NA1.1	IW	5	50	-6.9612	0.0004
2019__normal	B1.2	RW	1	50	-8.6881	0.0001
2019__normal	B1.3	RW	1	50	-8.6581	0.0001
2019__normal	NA1.2	RW	1	50	-8.5822	0.0001
2019__normal	R1.3	RW	1	50	-8.5807	0.0001
2019__normal	NA1.1	RW	1	50	-8.5637	0.0001
2019__normal	B1.1	RW	1	50	-8.5630	0.0001
2019__normal	R1.4	RW	1	50	-8.5629	0.0001

Table 10: Robustness to volatility-proxy choice: R1.1 reuses NA1.2 forecasts evaluated against $|r_t|$ (vol scale). Same QLIKE / RMSE definitions; forecasts taken as $\sqrt{\hat{\sigma}_t^2}$ for unit consistency.

robustness	scheme	horizon	n_obs	QLIKE	RMSE
R1.1 abs_return target, vol scale (reuses NA1.2 forecasts)	RW	1	252	-3.6695	0.0104
R1.1 abs_return target, vol scale (reuses NA1.2 forecasts)	RW	5	252	-2.0541	0.0303
R1.1 abs_return target, vol scale (reuses NA1.2 forecasts)	IW	1	252	-3.6643	0.0105
R1.1 abs_return target, vol scale (reuses NA1.2 forecasts)	IW	5	252	-2.0654	0.0300

Table 11: Forecast combination test: $(1-w) \cdot \text{B1.2} + w \cdot \text{NA1.2}$. w^* near zero indicates NA1.2 carries no orthogonal information vs GJR.

scheme	horizon	model_a	model_b	n_obs	QLIKE_a_alone	best_w	QLIKE_combined	Δ QLIKE
RW	1	B1.2	NA1.2	252	-7.7576	0.0000	-7.7576	0.0000
RW	5	B1.2	NA1.2	252	-6.3667	0.0000	-6.3667	0.0000
IW	1	B1.2	NA1.2	252	-7.7581	0.0000	-7.7581	0.0000
IW	5	B1.2	NA1.2	252	-6.3653	0.0000	-6.3653	0.0000

Table 12: Economic significance of the news effect under fitted NA1.2 (NA-GARCH-asym). $f(P, N)$ is the news-scaling factor; values at the four corners of the $(P, N) \in [0, 1] \times [-1, 0]$ support box. `range_pct_baseline` expresses the $f_{\max} - f_{\min}$ range as a percentage of the no-news baseline $f(0, 0)$.

quantity	value
$f(P=0, N=0)$	0.9829
$f(P=1, N=0)$ max-dov	0.9849
$f(P=0, N=-1)$ max-haw	1.0247
$f(P=1, N=-1)$ max-mix	1.0267
max	1.0267
min	0.9829
range_pct_baseline	4.4567

Table 13: Appendix: alternative input constructions for the news term in NA-GARCH-asym. R1.8 uses MPS-vs-intra-period-baseline surprises with 5-day exponential decay; R1.9 and R1.10 use Bernoth-style residualization of P_t and N_t on financial-state controls with and without AR(2) lags of the stance series, respectively. $\Delta\ell$ is logL minus logL under NA1.2 (raw levels) on the same observations. All three alternative constructions underperform raw levels.

model	description	dist	n_obs	loglik	aic	delta_ll_vs_b22	delta_aic_vs_b22
R1.8	MPS surprise vs intra-period baseline, 5-day exp decay	studentst	2754	8195.2170	-16374.4340	-0.0720	NaN
R1.9	Bernoth surprises, with AR(2) lags	studentst	2752	8188.2260	-16360.4520	-1.4490	2.8980
R1.10	Bernoth surprises, no AR lags	studentst	2754	8193.6530	-16371.3060	-1.6360	3.2710